Vivek Raj Singh

Bhopal, Madhya Pradesh

+91-9119672806 | rajvivek22102004@gmail.com | GitHub

Summary

Undergraduate passionate about quantitative finance and computational modeling. I enjoy building pricing engines, running Monte Carlo simulations, and thinking about how stochastic processes can explain real-world risk. Currently deepening my skills in derivatives pricing, time-series econometrics, and algorithmic trading. Published researcher with FOSSEE/IIT Bombay.

EDUCATION

VELLORE INSTITUTE OF TECHNOLOGY **2023 – 2027**

B.Tech - Computer Science and Engineering (Cloud Computing) – CGPA: 8.60 *Bhopal, Madhya Pradesh*

Sunbeam English School, Bhagwanpur **2020 – 2022**

Higher Secondary: 75.2% (2021–2022) | Matriculation: 93.2% (2019–2020) *Varanasi, U.P.*

TECHNICAL SKILLS

- Languages: Python, Scilab, Rust, C++, SQL
- Quant Libraries: NumPy, SciPy, statsmodels, pandas, Matplotlib
- ML / Deep Learning: PyTorch, scikit-learn, TensorFlow/Keras
- Tools & Infra: Git/GitHub, Docker, FastAPI, MongoDB, LaTeX
- Core Concepts: Stochastic Calculus, Monte Carlo Methods, Options Pricing, VaR/CVaR, Time-Series Analysis, Credit Risk

PROJECTS

H2O Automated Machine Learning Pipeline | *Python, H2O.ai, Deep Learning, Random Forest*

– Engineered a multi-model distributed machine learning pipeline utilizing the H2O.ai framework for high-performance classification and predictive modeling.
– Implemented automated data preprocessing routines including mean imputation, categorical encoding, and stratified dataset partitioning for robust validation.
– Trained and bench-marked multiple algorithms including Random Forests and Simple Neural Networks, optimizing a Tuned Deep Learning model to achieve a superior logloss of 0.0915.

Stochastic Risk Quantification Engine | *PyTorch, statsmodels, Monte Carlo, FastAPI*

– Modeled hardware degradation as a mean-reverting stochastic process; synthesized 730 days of statistically rigorous telemetry data with realistic drift and volatility clustering.
– Designed a weighted ensemble forecasting pipeline combining custom PyTorch LSTMs (Cosine Annealing LR, Huber Loss) with ARIMA for robust multi-horizon prediction.
– Built a quantitative risk engine running 10,000 Monte Carlo simulations to compute tail-risk metrics (VaR_{95}, $CVaR_{95}$); deployed Isolation Forests for anomaly detection.
– Exposed inference via a thread-safe FastAPI service with serialized model bundles for real-time risk scoring.

Quantitative Portfolio Analytics | *Excel, Markowitz Optimization, Monte Carlo*

– Built a Markowitz mean-variance optimizer for a 2-asset portfolio (SBIN + ICICI Bank): computed covariance matrices, generated 50 weight combinations, and plotted the Efficient Frontier.
– Ran 25-path Monte Carlo simulations on NIFTY 50 calibrated to historical volatility to estimate forward price distributions and Value at Risk.
– Constructed Bollinger Bands (20-day MA $\pm$ 2σ) on NIFTY 50 closing prices to identify overbought/oversold conditions and mean-reversion signals.

CONTRIBUTIONS

Simulating and Pricing Options under the Heston Stochastic Volatility Model **Apr 2026**

– Published as an official FOSSEE Case Study (IIT Bombay) on the Scilab national portal; independently peer-reviewed.
– Implemented dual-validation architecture: analytical Fourier inversion pricing cross-validated against a Monte Carlo engine with Antithetic Variates achieving ~65% variance reduction.
– Reproduced the empirical volatility smile via implied vol extraction and rendered a full 3D option price surface over a 13×10 (strike, maturity) grid.

CERTIFICATIONS

– Pursuing Executive Programme in Algorithmic Trading (EPAT) – quantitative finance, statistical arbitrage, time-series modeling, and algorithmic trading systems.
– FOSSEE Summer Fellowship 2026 (IIT Bombay) – Selected for the Scilab Case Study programme in computational finance.